



Cloud Security with AWS IAM



o Thato Ntoahae

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": "ec2:*",  
7       "Resource": "*",  
8       "Condition": {  
9         "StringEquals": {  
10          "ec2:ResourceTag/Env": "development"  
11        }  
12      }  
13    },  
14    {  
15      "Effect": "Allow",  
16      "Action": "ec2:Describe*",  
17      "Resource": "*"   
18    },  
19    {  
20      "Effect": "Deny",  
21      "Action": [  
22        "ec2:DeleteTags",  
23        "ec2:CreateTags"  
24      ],  
25      "Resource": "*"   
26    }  
27  ]  
28 }  
29
```

Introducing Today's Project!

Project overview

In this project, I will demonstrate how to use ec2 instance and iam... I'm doing this project to learn...IAM and how control access to a internet computer or cloud.

Tools and concepts

Services I used were...ec2 instance and IAM. Key concepts I learnt include...creating two instances and adding a IAM policy to manage permissions for users and user groups that what learnt.

Project reflection

This project took me approximately...2h The most challenging part was...creating the iam policy. It was most rewarding to...see my instances were running and IAM did exactly what i wanted it to do.

Tags

What I did in this step

In this step, I will... create two ec2 instances because...i need to boost computing power

Understanding tags

Tags are labels or keywords assigned to resources (like files, cloud instances, or code) to help categorize, organize, and identify them. They make it easier to search, filter, and manage items, especially in large systems.

My tag configuration

The tag I've used on my EC2 instances is called...nextwork-prod.....The value I've assigned for my instances are...production the other instance is called nextwork-dev....and the value is development

Key Info	Value Info	Resource types Info	Remove
Name	nextwork-dev-that	Select resource types	
		Instances	

Key Info	Value Info	Resource types Info	Remove
Env	Development	Select resource types	
		Instances	

IAM Policies

What I did in this step

In this step, I will...create an IAM policy because...i want to give the intern access to the development instance.

Understanding IAM policies

An IAM policy is a rule for who can do what with your AWS resources. It's all about giving permissions to IAM users, groups, or roles, saying what they can or can't do on certain resources, and when those rules kick in.

The policy I set up

For this project, I've set up a policy using...JSON

Policy effect

I've created a policy that...allows and denies any action that will be made in the EC2 instance.

Understanding Effect, Action, and Resource

The Effect, Action, and Resource attributes of a JSON policy means...the policy allows some actions (like starting, stopping, and describing EC2 instances) for instances tagged with "Env = development" while denying the ability to create or delete tags for all instances.

My JSON Policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": "ec2:*",  
7       "Resource": "*",  
8       "Condition": {  
9         "StringEquals": {  
10          "ec2:ResourceTag/Env": "development"  
11        }  
12      }  
13    },  
14    {  
15      "Effect": "Allow",  
16      "Action": "ec2:Describe*",  
17      "Resource": "*"   
18    },  
19    {  
20      "Effect": "Deny",  
21      "Action": [  
22        "ec2:DeleteTags",  
23        "ec2:CreateTags"  
24      ],  
25      "Resource": "*"   
26    }  
27  ]  
28 }  
29
```

Account Alias

What I did in this step

In this step, I will...Create an AWS Account Alias because...users can have a simple way of accessing the account.

Understanding account aliases

An account alias is...a friendly name for your AWS account that you can use instead of your account ID (which is usually a bunch of digits) to sign in to the AWS Management Console

Setting up my account alias

Creating an account alias took me...2min Now, my new AWS console sign-in URL is...<https://638954280586.signin.aws.amazon.com/console>

New sign-in URL

<https://nextwork-alias-thato.signin.aws.amazon.com/console>

 IAM users will still be able to use the default URL containing the AWS account ID.

IAM Users and User Groups

What I did in this step

In this step, I will...create iam users and groups because...its going to allow users to access what they need.

Understanding user groups

IAM user groups are...a collection/folder of IAM users. It allows you to manage permissions for all the users in your group at the same time by attaching policies to the group rather than individual users.

Attaching policies to user groups

I attached the policy I created to this user group, which means...i can manage permissions for all the users in the group rather than individuals.

Understanding IAM users

IAM users are... are the people that will get access to your resources/AWS account

Logging in as an IAM User

Sharing sign-in details

The first way is...using a password manager and the second way is to share the password via a secure channel

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed... that it took me straight to the console as a new user. This was because...i gave access to users for the development instance and now people can access it.

Console sign-in details

[Email sign-in instructions ↗](#)

Console sign-in URL

<https://638954280586.signin.aws.amazon.com/console>

User name

[nextwork-dev-thato](#)

Console password

[***** Show](#)

Testing IAM Policies

What I did in this step

In this step, I will...test the interns access because...we want to make sure they have the right access to the development instance.

Testing policy actions

I tested my JSON IAM policy by...trying to stop the instances but an error appeared deny access and then i went to my ec2 console to stop the instances and it allowed me.

Stopping the production instance

When I tried to stop the production instance...it failed to stop the instance. This was because...it was not authorized to do so.

 Failed to stop the instance i-0085b1040f0e9d073 Diagnose with Amazon Q

You are not authorized to perform this operation. User: arn:aws:iam::638954280586:user/nextwork-dev-thato is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:us-east-1:638954280586:instance/i-0085b1040f0e9d073 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message:
F7QwZD_XfcttbMWqN9jYijA52agA7NFzO84GkvQTkwb3IBKRmZHkFvgJLCmhlkBD8X2r_0QqRLCw64hIJRJBxiECy
NF4q6vsGfG6ApIjACTzKW7SEgoiky9CvoBGoBNmpKP7VhN37jiw6b7sliyQka-
duPt1RspJQEWNPARDM4dCN_9snK5sT_t_ZMZ_TPyUXPzzaQaXuMiKxNo1tEw1aPcy561CKydomuzOIOP19bCDqs
IGn99VlqeM6JXRAt_YB2tjvK8wbPO8-
0Xe5sORc4OJeNur4Enp7wWHODCKw9FBojMpOILNxnAaJeobZoyV0Qgxf4uBu3cmETMBZWzsE06aN6BTabaUJM1
a5bRVmWFWoVN2M8vMmb-zigU9goQ122Cn0JtnnlW0F0owlcFw7o2bZs-
dOB0GDdDAuSD3OYsFKXHNiqTS_zZQd2KA_rKG2YKnghe4rM0in28sHZ4ccStwOvJs16w7jAo7AHMxtGVfPQKX
7gEKVZebhh7Tda-

Stopping the development instance

Next, when I tried to stop the development instance...an error showed up. This was because...the policy did not give any permissions for stopping any instances.

 Successfully initiated stopping of i-042f2fe574ace81a5

IAM Policy Simulator

To extend my project, I'm going to...Use the IAM Policy Simulator. I'm doing this because...it's best practice to run these tests in another way.

Understanding the IAM Policy Simulator

The IAM Policy Simulator is...another way to run tests. It's useful for...running this instances in another way.

How I used the simulator

I set up a simulation for...nextwork-dev. The results were...denied. I had to adjust...the tags so that it allows the permissions granted.

	Service	Action	Resource Type	Simulation Resource	Permission
▶	Amazon EC2	DeleteTags	not required	*	denied 1 matching statements.
▶	Amazon EC2	StopInstances	instance	*	allowed 1 matching statements.



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